

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Fire and Explosion Philosophy
Document Number	EA-BFPCL-GPMC-GGPL-FEED-THS-PHY-001
Document Revision	R02
Document Status	Issued for Review
Originator	Ekene Okoye
Security Classification	Restricted
Issue Date	14-Jul-22

Fire and Explosion Philosophy

R02	14.07.22	Re-issued for Review	E. Okoye	E. Maduabuchi	M. Ekpunobi	
R01	15.05.22	Issued for Review	E. Okoye	E. Maduabuchi	K. Yusuff	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	15-05-22	Issued for Review
R02	14-07-22	Re-issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-THS-PHY-001

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Project Information, Terms, Abbreviations, and Definitions

Project Information – General

1.0	Employer	Brass Fertilizer & Petrochemical Company Limited (BFPCL)
2.0	PMC	Tata Consulting Engineers Limited (TCE)
3.0	EPCC Contractor	China Tianchen Engineering Corporation (TCC)
4.0	Project Title	Gas Processing and Methanol Complex (GPMC)
5.0	Location	Akabel, Brass Island: • 85km South West of Port Harcourt, • 70km South of Yenagoa and • 90km West of Bonny Island.
5.1	State	Bayelsa State, Nigeria
5.2	Nearest Air Port, Sea Port. Railway Station	Port Harcourt (PHC), 85km Bayelsa Airport (Under Construction)
6.0	Standard Process Condition	15°C and 1.01325 bar
7.0	Design Life	30 years

Definition of Terms

Terms	Definitions
BFPCL	Brass Fertilizer & Petrochemical Limited.
SPDC	Shell Petroleum Development Company Nig. Limited
Employer	Party (Usually BFPCL) that initiates the project and ultimately pays for its design and construction. The Employer will generally specify the technical requirements. The Employer may also include an agent or consultant authorized to act for and on behalf of, the Employer.
Engineer	The party that carries out the FEED Engineering of the project.
Manufacturer/Supplier	Party, which manufactures or supplies equipment and services to perform the duties specified by the Contractor.
Shall	The word 'shall' indicate a mandatory requirement.
Should	The word 'should' indicate a recommendation.
Measuring Range	Range defined by LRL and URL within which the limits of uncertainty of the measuring instrument are specified (the capability of the instrument).
On-Line Instruments	Instruments connected to process and utility lines or equipment via primary isolation valves.

Project	Front End Engineering Design for Gas Gathering Pipelines (Phase1)
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Abbreviations

Abbreviation	Meaning
ALARP	As Low As Reasonably Practicable
CPF	Central Processing Facility
ESDV	Emergency Shutdown Valve
FAR	Field Auxiliary Room
HSSE	Health Safety Environment and Security
PFP	Passive Fire Protection

Site and Environmental Conditions*

1.	Air Temperature	Average daytime 32°C; rainy season 28°C; night time 23°C
2.	Weather	Dry Season (Harmattan); November – January; the atmosphere is laden with extremely fine dust (5 microns particle size) Wet Season: heavy rainfall, two peaks, August- 245mm, September- 331mm
3.	Relative Humidity	Mean 81%; Minimum 69% & Maximum 88%
4.	Wind Speed and Direction	Monthly mean 8.5km/h; varies from 6.2km/h to 9.6km/h The south-westerly wind is 60% predominant during the wet season (March - October). The north-easterly wind is 32% predominant during the dry season (November– February)
5.	Seismicity	The project area is classified as Seismic Zone 0

*Data from ESIA Report

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opubene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

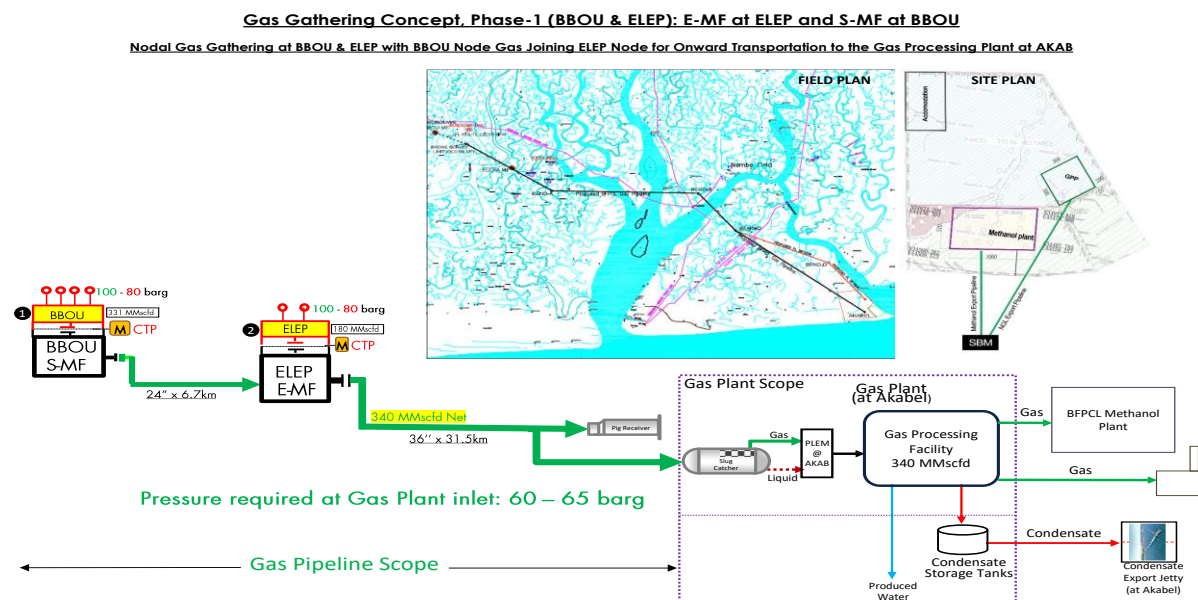


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of this document is to establish the fire and explosion approach and principles that will guide the design of the surface facilities for the Project, to ensure that residual HSSE risk is as low as reasonably practicable (ALARP). This is a live document that will be updated as the project progresses.

In the engineering design phase, there is considerable scope to remove or minimise hazards. This can be achieved in a number of ways. In order of preference:

1. Removal of hazards
2. Reducing the probability of hazardous events occurring
3. Minimising the risk of escalation should a hazardous event occur.

Protection against fire and explosion primarily concentrates on minimising the risk of escalation through recovering from loss of containment and the consequences thereof. This will be achieved using the following guiding principles:

- Detect any potential, or actual, loss of containment
- Isolate the facility inventory
- Minimise the duration of any event by reducing pressure and inventory
- Provide appropriate separation between equipment to minimise escalation
- Minimise the risk of escalation by reducing the probability of ignition
- Extinguish any resultant fire
- Reduce the effects of any resultant explosions
- Ensure emergency power and communication facilities
- Allow escape, evacuation and rescue of personnel.

This document establishes the following design philosophies for the project in support of the above principles:

Plant Layout, Process Containment, Bunding & Drainage

1. Ignition Control
2. Explosion Mitigation
3. Passive Fire Protection
4. Active Fire Protection
5. Access and Escape

The following further design philosophies have been outlined in HSSE Design Philosophy of the project, again in support of the above principles:

- Process Safeguarding Philosophy
- Drainage Philosophy
- Fire and Gas Detection and Protection Philosophy
- Shutdown Philosophy
- Relief, Blowdown and Venting / Flaring Philosophy

The systems will generally be designed in accordance with Shell DEP's, except where noted in this

document. Deviations to DEP's will be controlled by deviation requests, and approved by the relevant technical authority. However, the key DEP's are mainly related to the fire hazard assessment process as opposed to specifying specific requirements and thus deviation requests are considered unlikely.

The fire and explosion philosophies have been developed to provide a framework for FEED to ensure that the risks to personnel from the operation of the plant are ALARP and to ensure there is an appropriate level of asset protection.

1.3. Scope

The scope of this philosophy document encompasses those issues that affect the design of the facilities included within the Project scope of work, which includes:

- Satellite Manifold at BBOU complete with all Process, Utility & Control Infrastructure, however with provision for PIG receiver in phase 2 production from OPUG.
- Evacuation Manifold at ELEM complete with all Process, Utility & Control Infrastructure, however with provision for PIG receiver in phase 2 production from NUNR.
- Pipeline from BBOU to ELEM
- Pipeline from ELEM to AKAB (up to Inlet Flange of Slug Catcher)
- PIG Launchers at BBOU and ELEM Manifold
- PIG Receivers at ELEM and AKAB
- Manifold control system also to be provided. Control Room required at each Manifold to house the Manifold Control system & Telecom system
- Cabinet Room to house the electrical MCC.
- UPS Power for Control System, associated Civil, Mechanical, Pipeline, Electrical, Instrumentation & Control, Telecommunication, and all other supporting infrastructure.

The Scope of the FEED Engineering covers the provision of all necessary information, technical drawings, philosophies, specifications, design data, listing of codes and standards required for the detailed design and construction of the Multiphase pipeline system.

2. Regulations, Codes And Standards

The latest versions of the regulations, codes, and standards and extant amendment or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

2.1. Nigerian Standards & Regulations

Document Number	Title
(EGASPIN 2018)	Environmental Guidelines & Standards for the Petroleum Industry in Nigeria
MOSR 1997	Minerals Oil & Safety regulation
DPR October 2020	DPR Safety Case Guidelines for Oil & Gas Facilities in Nigeria

2.2. Shell Design and Engineering Practices (DEPs)

Applicable DEPs, supplements and MESC SPE Codes are as referenced in various discipline dossiers, some of which are listed below.

Document Number	Title
DEP 80.00.10.11	Gen Layout of Onshore Facilities
DEP 30.00.60.10-GEN	Human Factors Engineering in Projects
DEP 33.64.10.10	Electrical Design
DEP 80.45.10.10	Pressure Relief, Emergency Depressurising, Flare and Vent Systems
DEP 80.47.10.30-Gen	Assessment of Fire Safety of Onshore Installations
DEP 32.30.20.11	Fire, Gas and Smoke Detection
DEP 80.00.10.10	Area Classification (Amendments/Supplements to IP 15)
DEP 80.47.10.31-Gen:	Active Fire Protection Systems and Equipment for Onshore Facilities
DEP 80.47.10.32-Gen:	Portable and Mobile Equipment for Fire-Fighting

2.3. International Codes And Standard

Discipline	Document Number	Title
	Part 15, 3rd Edition	Area Classification Code for Installations Handling Flammable Fluids, Institute of Petroleum Model Code of Safe Practice
	IEC 60079	Explosive Atmospheres – Part 0: Equipment – General Requirements
	NFPA 101	National Fire Protection Association Code for Safety

3. Plant Layout, Process Containment, Bunding & Drainage

3.1. Layout

The following philosophy shall be adopted:

Layout Factor	Rationale
Close coupling of vessels in same process stream.	Minimises both piping to build, inventory and land take. However, this requirement is to be balanced with segregating plant to minimise fire and explosion escalation.
Segregation of process trains.	Minimises asset and production loss from fire by minimising the potential for escalation and damage.
Segregate essential utilities (generation/fuel gas system/ instrument air) from process facilities.	Minimises asset and production loss from fire and explosion.
Provide adequate access for fire fighting.	Allow swift response of Fire Brigade.
Locate inlet and outlet pipeline ESDV's at position out with fire areas.	Effectively isolate hydrocarbon inventory so that the outboard pipelines cannot be subject to harmful thermal radiation.
Locate the site entrances and any "safe" facilities (switch room / weather shelters) upwind with respect to prevailing wind. Locate drains sump and vents/flares downwind.	Minimise gas cloud interaction potential and improves egress reliability. Ensures hydrocarbon drains/pool fires run away from manned areas and the site egress.
Non-hazardous utilities to be located between the hazardous areas and control room/accommodation if provided.	Act as a barrier protecting the control room and the accommodation from hazardous events in the process areas.
All process equipment to be out in open in naturally ventilated areas whenever practicable.	Minimises the potential for gas build up and minimises explosion overpressure potential.
Ensure no perimeter trees have potential to fall on to the plant and create damage.	Eliminate damage from high probability event.
Ensure no perimeter trees are close to vents/flare(s).	Eliminate trees/jungle catching fire.
Go for longer thinner layout as opposed to square layout.	Maximises separation distances.

Provide material handling and suitable crane access.	To allow good maintenance access.
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3.2. Process Containment

One of the more important areas to mitigate risk is in the area of process containment. The philosophy will be to keep the plant as simple and robust as possible and minimise the potential for loss of containment events. The number of valves (hence flanges) will be minimised, i.e. isolation valves will be provided at the boundaries of trains and additional valves will be avoided whenever possible.

The need for small bore process piping shall be critically examined and shall not be used unless strictly required. If such piping is required, it shall be designed properly and protected against mechanical failure as a result of vibration, corrosion or over-pressuring of blocked-in pipe sections.

A process simplification review will held be during FEED.

3.3. Bunding & Drainage

The process footprint will be minimised. Concrete slabs or drip trays will (only) be provided under process equipment with significant liquid spill potential. The area with liquid spill potential shall be kerbed and routed to the open drains. The remainder of the plant shall be drained to the normal rainwater/storm drains with no oil recovery (valving will be provided to allow manual routing to open drains for treatment, if required). A drainage philosophy is contained in the HSSE Design Philosophy of the project.

No fire water will be required unless justified by fire and explosion risk assessment

Consideration will be given to providing simple roofs (with additional side cladding) to cover weather shelters and storm water from the shelters will be routed directly to the swamp. Roofs and side panels may increase the potential for the accumulation of gas clouds and hence increase the ignition potential, and could increase potential explosion over-pressures. During FEED the negative impact of these shall be reviewed against their benefits and the ALARP risk option selected.

4. Ignition Control

All remote platforms will be categorised in Hazardous Area Classification based on the Institute of Petroleum's Model Code of Safe Practice IP part 15 Edition.

The type of equipment being installed will predominantly create Zone II hazardous areas. There may be some small, localised Zone 0 hazardous areas around vent discharges. All electrical equipment out on site will be rated for use in Zone II hazardous areas and be installed outwith any Zone I or Zone 0 hazardous areas.

Power generation equipment where provided will be located in non-hazardous areas whenever practicable. The combustion and ventilation air inlets will be located as far as is reasonably practicable and at least 1 metre outside a hazardous area.

Lightning protection shall be provided on tall structures where appropriate. The lightning protection shall be designed to cover the entire manifold. Field transmitters and other sensitive items will be protected additionally with suitable earthing.

5. Explosion Mitigation

Plant located in the open or with simple roof structures will have minimal gas cloud formation potential, thus explosion overpressures and associated explosion risks will be low. Modularised plant with plated roofs and louvered side panels will have an increased potential for gas cloud formation and will create an increased risk from explosion overpressure. The potential for escalation and asset damage will be assessed, and if necessary, measures to reduce the risk will be selected using a risk-based approach.

Equipment rooms for Electrical and Control/Safeguarding equipment will be located at the periphery of the process and protected with a blast wall to minimise their exposure to blast overpressure. The rooms shall be designed according to the maximum blast overpressure they could experience. Guard accommodation and offices where provided should be located in an area free from explosion hazards or should be designed to be blast resistant as appropriate.

6. Passive Fire Protection

The main (or even sole) objective of fireproofing of steel structures is to prevent the escalation of fires to an acceptable level by providing a temporary protection until full fire-fighting capabilities can be deployed. If active fire protection systems also covering the area where the equipment is located are not expected to give timely and adequate protection, passive fire protection shall be provided. Passive fire protection (PFP) shall be used in exceptional cases where activation of active systems is expected to be too slow or inadequate to avoid escalation.

Certain equipment which must continue to operate during a fire, such as remote-operated emergency shut-off and depressurising valves and actuators or critical electrical instrument cables, may need fire protection to stay operable for a defined period of time

To minimise the occurrence of under-lagging corrosion and to avoid additional cost, passive fire protection will only be installed where fire and explosion risk assessment justifies the need.

7. Active Fire Protection

Active fire protection may be required if justified by fire and explosion risk assessment (FERA), so that the potential fire risk is managed to as low as is reasonably practicable. The remote manifold stations will be operated as unmanned facilities but will be attended to by Maintenance and Operations

The Fire water system if required shall be designed in accordance with DEP 80.47.10.30 and DEP 80.47.10.31.

8. Access & Escape

Good access to all process equipment shall be provided, including maintenance access for equipment change out.

Two diverse means of escape shall be provided from all process areas whenever reasonably practicable.

9. Pipelines

No specific fire or explosion protection will be provided for the pipelines. The pipelines shall be welded unless they are within manifolds or the CPF at Akabel.